

Functional & Performance Testing Template

Model Performance Test

Date	02 October 2026
Team ID	SWTID-2026-1665
Project Name	EduGenie: Google Gemini Powered Learning Assistant
Maximum Marks	

Test Scenarios & Results

Test Case ID	Scenario (What to test)	Test Steps (How to test)	Expected Result	Actual Result	Pass/Fail
FT-01	Text Input Validation (topic, question, passage)	Leave the text area empty, enter symbols only, then enter a valid question/topic (e.g., "Which is the largest ocean?") and submit	Valid text is accepted and processed; empty or invalid input shows a clear error message	To be recorded after execution	Pending
FT-02	Input Length Validation (very short and very long passages)	Submit a one-word input, a normal paragraph, and a very long passage for Summary/Quiz	Valid lengths are accepted; out-of-range input shows an error or is handled gracefully without crashing	To be recorded after execution	Pending
FT-03	Content Generation (Q&A, Explain, Quiz, Summary, Learning Path)	Select each task in the dropdown (Q&A, Explain, Quiz, Summary, Recommend Path), enter a topic (e.g., Pythagoras Theorem, SQL) and click Submit	Q&A gives a concise answer; Explain gives a simplified explanation; Quiz gives 3 MCQs with 4 options and the correct answer; Summary is concise; Path is beginner-to-advanced with resources	To be recorded after execution	Pending
FT-04	API Connection Check (Gemini API and local explanation model)	Verify the Gemini API key is configured, call /qa, /quiz, /summarize and /learn/recommendations, and call /explain to check the local LaMini-Flan-T5 model loads	All endpoints respond successfully with no authentication or model-loading errors	To be recorded after execution	Pending
PT-01	Response Time Test	Use a timer to measure the time from clicking Submit to the result appearing for each task	Content is generated in under 3 seconds	To be recorded after execution	Pending
PT-02	API Speed Test (concurrent requests)	Send multiple requests to the endpoints at the same time (e.g., several Q&A and Quiz calls together)	API does not slow down noticeably and all requests return valid responses	To be recorded after execution	Pending
PT-03	Large Passage Load Test (long educational text)	Submit multiple long passages one after another for summarization and quiz generation and monitor processing	Processing works smoothly without crashing or freezing the application	To be recorded after execution	Pending